Day 03 of 60

The Exact Path To Learn ML From Scratch

No fluff. Here's the lean, practical 6-month roadmap I'd follow if I had to start over today.

📷 6 Slides | ⏱ 3 Min Read | ⚡ Roadmap

Visual Roadmap
The step-by-step path from basic Python to model deployment.

Lean Tool Stack
Skip the bloat. Only learn the tools that matter in production.

Detours to Avoid
5 common beginner traps that will waste months of your time.

THE VISUAL PATH

Follow this exact sequence. Do not skip steps.

1. Foundations (Code + Data)

Master Python, SQL, and Pandas. If you can't clean and manipulate data, you can't do Machine Learning. Learn basic linear algebra and statistics on the side.

2. Core Machine Learning

Dive into Scikit-Learn. Understand Supervised learning (Linear/Logistic Regression, Random Forests, XGBoost) and Unsupervised learning (K-Means). Focus on model evaluation.

3. Deep Learning

Pick PyTorch. Build Neural Networks from scratch. Move on to CNNs (Computer Vision) and Transformers (NLP/LLMs) using HuggingFace.

4. MLOps & Deployment

A model in a notebook is useless. Learn how to wrap models in APIs (FastAPI), containerize them (Docker), and deploy to the cloud (AWS/GCP/Render).

TIME ESTIMATES

A realistic 6-month breakdown assuming 10-15 hours per week.

Months 1-2

The Setup

Focus entirely on getting fluent with Python syntax and wrangling messy datasets in Pandas. If you skip this, everything else will be frustrating.

Months 3-4

The Core Intuition

Train classical models on tabular data. Learn how algorithms learn, how to detect overfitting, and how to tune hyperparameters properly.

Months 5-6

The Advanced Shift

Transition to Deep Learning architectures and MLOps. Your goal is to build an end-to-end project and deploy it so others can interact with it via API.

💡 **CONSISTENCY OVER INTENSITY**

Don't try to cram this into 30 days. Your brain needs time to build intuition for how algorithms behave on different datasets. Give it 6 months.

THE LEAN TOOL STACK

Stop learning tools you won't use. This is the 80/20 stack used in production today.

Data Prep	`python` `pandas` `numpy` `sql`
Core ML	`scikit-learn` `xgboost` `matplotlib`
Deep Learning	`pytorch` `huggingface` `transformers`
Deployment	`fastapi` `docker` `git` `aws/gcp`

5 DETOURS TO AVOID

Avoid these beginner traps that will derail your progress.

01

Starting directly with Deep Learning (LLMs)

Skipping traditional ML guarantees you won't understand model evaluation, overfitting, or feature engineering. Build foundations first.

02

Memorizing Math Proofs

You don't need a PhD in math to build models. Understand the intuition behind the math, but let the code do the actual calculations.

03

Framework Hopping

"Should I learn TensorFlow or PyTorch?" Just pick PyTorch and stick with it. Switching frameworks wastes time you could spend learning concepts.

04

Ignoring Data Cleaning

80% of ML is data prep. If you only train models on perfect Kaggle datasets, you will fail in the real world where data is messy.

05

Tutorial Hell

Watching 100 hours of video without writing code is useless. Build a tiny, messy, imperfect project as soon as possible to solidify knowledge.

WHAT'S NEXT

03

✓ Day 03 Complete

Tomorrow on Day 04:

The Three Main Types of ML

Supervised vs. Unsupervised vs. Reinforcement — visual breakdown

Follow for the full 60-day ML series

Vinod Bavage

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